SKILL DEVELOPMENT CELL

Day 2 – Data Analytics Hackathon

Problem Statement: Data to Decisions – IPL Edition

“While teams in Indian Premier League’26 are busy winning matches, it’s your turn to win with data.” “If you’ve been shouting ‘What a shot!’ during the IPL, now it’s time to explain it with data.”

In today’s sports ecosystem, data plays a crucial role in decision-making.
Leagues like the Indian Premier League (**IPL**) generate massive ball-by-ball datasets, capturing every detail of the game.

However, raw data alone is not useful unless it is transformed into meaningful insights.

Teams, analysts, and strategists rely on data analytics to:

- Understand player performance
- Optimize match strategies
- Identify winning patterns

Despite the availability of rich datasets, extracting actionable insights remains a challenge.

Now it’s your turn to turn IPL’s ball-by-ball data into insights that drive decisions.
Data set : IPL RAW DATA SET

Challenge

Design and develop a data analysis solution using Kaggle Notebooks that transforms raw IPL ball-by-ball data into meaningful insights through visualization and storytelling.

Objective

Analyze the dataset and build a structured data story that enables:

- Identification of player performance trends
- Strategic insights for teams
- Understanding of match dynamics
- Clear, data-driven conclusions

Core Requirements (Minimum)

🔹 1. Exploratory Data Analysis (EDA)

Participants should:

- Clean and preprocess the dataset
- Perform structured data exploration
- Identify key patterns and trends
- Innovative Ideas(**Bonus**)

🔹 2. Player Performance Analysis

Participants should analyze:

- Most consistent batsmen across seasons
- Performance in high-pressure situations (death overs)
- Most economical bowlers
- Innovative Analysis Ideas(**Bonus**)

🔹 3. Team Strategy Insights

Participants should evaluate:

- Impact of toss on match outcomes
- Success rate of chasing vs defending
- Influence of venue on performance

🔹 4. Match Dynamics Analysis

Participants should explore:

- Run rate progression across innings
- Momentum-shifting overs
- Phase-wise performance:
    - Powerplay
    - Middle overs
    - Death overs

Optional Enhancements (Bonus)

- Advanced visualizations (interactive charts)
- Player comparison dashboards
- Insight storytelling with clear narratives
- Unique patterns or hidden trends

Technical Guidelines

Participants must use:

- **Platform:** Kaggle Notebook (mandatory)

You may use:

- Python
- Visualization libraries

Additional Requirements

- Focus on insights, not just plots
- Avoid unnecessary or repetitive visualizations
- Maintain a clear and logical storytelling flow
- Ensure code is clean and well-structured

Deliverables

Participants must submit:

- **Kaggle Notebook Link**-*clear explanations within the notebook in the form of comments*
- **Demo video of visualization** (*Once you finish, You must have to run the complete code from starting cell to end, showing the results in the screen-recording(**no voice needed**))*

Evaluation Criteria

- **Insight Quality — 30%**
- **Storytelling — 25%**
- **Visualization — 20%**
- **Technical Execution — 15%**
- **Innovation — 10%**

ALL THE BEST